EXERCISE 8

Physical Regimes of the Damped Harmonic Oscillator

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Mechanical Damping • Under/Critical/Overdamped

PHASE 1: STATEMENT & SETUP

> Consider the equation $m\ddot{x} + \mu\dot{x} + kx = 0$ for the deviation $x(t)$ of a point particle with mass m from its equilibrium position, subject to an elastic force (constant k) and drag force (parameter μ). Discuss how the form of the solutions changes depending on the value of μ relative to the critical damping $\mu_c = 2\sqrt{mk}$.

PHASE 2: FUNDAMENTAL FORMULATION

The auxiliary polynomial is $mr^2 + \mu r + k = 0$. The discriminant is $\Delta = \mu^2 - 4mk = \mu^2 - \mu_c^2$, where $\mu_c = 2\sqrt{mk}$. Roots are $r_{1,2} = -\frac{\mu}{2m} \pm \frac{\sqrt{\mu^2 - \mu_c^2}}{2m} = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2}$.

PHASE 3: STEP-BY-STEP ANALYTICAL DERIVATION

1. Underdamped Regime: $\mu < \mu_c = 2\sqrt{mk}$ ($\Delta < 0$)

The discriminant is negative: $\sqrt{\mu^2 - 4mk} = i\sqrt{4mk - \mu^2}$.

Roots are complex conjugates with negative real part: $r_{1,2} = -\gamma \pm i\omega_d$, where:

$$\gamma = \frac{\mu}{2m}, \qquad \omega_d = \frac{\sqrt{4mk - \mu^2}}{2m} = \sqrt{\omega_0^2 - \gamma^2}$$

General solution:

$$\mathbf{x(t) = e^{-\gamma t}\left(c_1 \cos \omega_d t + c_2 \sin \omega_d t\right) = M e^{-\gamma t} \cos(\omega_d t - \phi)}$$

Behavior: Decaying oscillatory motion within an exponential envelope $\pm Me^{-\gamma t}$, crossing zero infinitely many times.

2. Critically Damped Regime: $\mu = \mu_c = 2\sqrt{mk}$ ($\Delta = 0$)

The discriminant vanishes identically. The characteristic equation has a single repeated real root:

$$r_1 = r_2 = -\frac{\mu_c}{2m} = -\sqrt{\frac{k}{m}} = -\omega_0$$

General solution:

$$\mathbf{x(t) = (c_1 + c_2 t)e^{-\omega_0 t}}$$

Behavior: Non-oscillatory. Crosses equilibrium at most once. Represents the fastest monotonic return to equilibrium without overshoot.

3. Overdamped Regime: $\mu > \mu_c = 2\sqrt{mk}$ ($\Delta > 0$)

The discriminant is positive: $\sqrt{\mu^2 - 4mk} < \mu$. Both roots are distinct, real, and strictly negative:

$$r_1 = -\gamma + \sqrt{\gamma^2 - \omega_0^2} < 0, \qquad r_2 = -\gamma - \sqrt{\gamma^2 - \omega_0^2} < 0$$

General solution:

$$\mathbf{x(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}}$$

Behavior: Purely exponential sluggish decay. At long times, dominated by the slower decay mode $e^{r_1 t}$. Crosses zero at most once.

PHASE 4: MASTER SUMMARY & ENGINEERING IMPACT

Damping Regime	Damping Condition	Roots $r_{1,2}$	General Solution $x(t)$	Dynamic Behavior
Underdamped	$\mu < 2\sqrt{mk}$	$-\gamma \pm i\omega_d$	$Me^{-\gamma t}\cos(\omega_d t - \phi)$	Oscillatory decay, infinite crossings
Critically Damped	$\mu = 2\sqrt{mk}$	$-\omega_0$ (mult. 2)	$(c_1 + c_2 t)e^{-\omega_0 t}$	Fastest return to trim without overshoot
Overdamped	$\mu > 2\sqrt{mk}$	$r_2 < r_1 < 0$	$c_1 e^{r_1 t} + c_2 e^{r_2 t}$	Sluggish exponential relaxation, no oscillations